

PILOT – Strong Cross-Domain Invariants (Privilege Domain)

Objective:

Define deterministic, cross-domain constraints that must hold for privileged activity to be valid.

1. Authenticated Session Requirement

Any privileged action must be associated with a valid, active authenticated user session at the time of execution.

2. Identity–Privilege Consistency

A privileged action must be consistent with the user's identity state (e.g. group membership, role assignment) at that exact time.

3. Credential–Session Binding

Use of a privileged credential must map to a specific user session and must not occur independently of an authenticated identity context.

4. Endpoint Context Consistency

A privileged action must originate from a valid endpoint that is associated with the authenticated user and authorised for that action.

5. Process–Identity Alignment

Any process performing a privileged action must be attributable to the authenticated user and consistent with expected execution context.

6. Temporal Consistency

All elements of a privileged action (authentication, credential use, execution) must align in time within acceptable tolerances.

7. Access Path Validity

The path from identity → credential → system access must be continuous and valid, with no missing or inconsistent links.

8. Session Continuity

Privileged activity must occur within a continuous session context; orphaned or detached actions are invalid.

9. Role Transition Integrity

Any elevation of privilege must correspond to a valid and recorded change in role or access rights.

10. Cross-Domain Coherence

Identity, privilege, and endpoint states must be mutually consistent; contradictions across domains indicate invalid system state.

Bottom Line

These invariants define what must be true for privileged activity to be possible.

Violations indicate causally inconsistent system states, not merely suspicious behaviour.